

Summary of Enhancements and Design Decisions

1. Graphical User Interface Enhancements

The project implements a fully interactive JavaFX-based graphical user interface designed for clarity, usability, and real-time feedback.

- **Layered Map Rendering**
The map is rendered using separate layers for roads, junctions, traffic lights, and vehicles. This layered approach improves rendering performance and simplifies visual updates, especially during simulation steps.
- **Zooming and Panning**
Mouse-based zooming (scroll wheel) and panning (right-click drag) were implemented to allow users to explore large maps easily. A double-click resets the view to the default position.
- **Junction Visualization**
Junction polygons extracted from the SUMO network file are drawn to visually connect roads and represent intersection areas accurately, improving spatial understanding of the road network.

2. Traffic Light Visualization and Control

Traffic lights are visually represented directly on the map and synchronized with SUMO's internal state.

- **Live Color Updates**
Traffic light colors (green, yellow, red) update in real time based on the current SUMO phase state.
- **Interactive Selection**
Users can select traffic lights directly on the map. The selected traffic light is highlighted, and its current state is shown in the control panel.
- **Manual Phase Control**
A control button allows advancing the selected traffic light to the next phase, enabling experimentation with signal timing.

Design Decision:

Only one visual indicator per traffic light was used to avoid clutter and ensure scalability when adding more intersections.

3. Vehicle Visualization and Differentiation

Vehicles are rendered dynamically with clear visual distinctions based on their type.

- **Vehicle Shapes and Colors**
 - Cars: yellow circles
 - Buses: blue rectangles
 - Emergency vehicles: orange triangles
- **Tooltips and Interaction**

Hovering over a vehicle displays detailed information such as ID, speed, type, and current road segment.
- **Selection Highlighting**

Clicking a vehicle highlights it for easier tracking.

Design Decision:

Simple geometric shapes were chosen for vehicle types to maximize readability and rendering performance.

4. Vehicle Filtering and Grouping

The interface supports dynamic filtering and grouping of vehicles.

- **Show/Hide by Vehicle Type**

Users can toggle visibility of cars, buses, and emergency vehicles using checkboxes in the control panel.
- **Live Vehicle Counts**

The control panel displays real-time counts for each vehicle type, updating every simulation step.

Design Decision:

Filtering is applied at the rendering level rather than modifying the simulation itself, preserving simulation correctness while offering flexible visualization.

5. Simulation Control Features

The control panel provides direct access to simulation functionality.

- **Start / Stop Simulation**
- **Rush Hour Mode** to increase traffic density
- **Emergency Vehicle Injection**
- **Traffic Light Phase Control**

Simulation startup is handled in a background thread to prevent blocking the UI.

6. Statistics Collection and Export

A comprehensive statistics subsystem was implemented to analyze simulation behavior.

- **Collected Metrics**
 - Average speed
 - Minimum and maximum speed
 - Total vehicle count
 - Vehicle density per road segment
 - Vehicle positions per simulation step
- **Export Functionality**
 - CSV export for numerical analysis
 - PDF export for formatted reporting
 - Vehicle position logs for post-processing
- **Automatic Export on Exit**

All statistics are automatically exported when the application closes.

Design Decision:

Statistics were decoupled from the GUI using a dedicated controller and exporter interfaces, ensuring clean separation of concerns and extensibility.

7. Multi-Threading Design

The application uses multiple threads to ensure responsiveness.

- The JavaFX Application Thread handles all UI rendering and user interaction.
- A dedicated background thread (SUMO-Start-Thread) initializes the SUMO simulation.

Design Decision:

Long-running backend operations are isolated from the UI thread to maintain a smooth user experience.

8. Modularity and Maintainability

The project was structured into clear components:

- GUI (MapView, ControlPanel)
- Simulation backend (SimulationManager)
- Statistics and export modules
- Utility services (logging)

This modular design simplifies debugging, testing, and future extensions.

9. Overall Design Philosophy

The main design goals were:

- **Real-time feedback**
- **Visual clarity**
- **Ease of use**
- **Extensibility**
- **Robust separation between UI, simulation logic, and data analysis**

The resulting system provides both an intuitive user experience and a solid technical foundation for further development.

The system was developed collaboratively, with responsibilities divided across user interface development, simulation backend integration, and statistical analysis.